INDUS UNIVERSITY

Monthly Progress Report - Software Group Project

(Report: 3)

Academic Year: 2026-27 Month: July

Project Title: House price prediction using AI/ML

Project Domain: AI/ML

Group Number: 50 Project Guide: Dhwani Mam

Student Details:

Enrolment No.	Student Name	Contribution	Signature
IU2341050103	Ravi Patel	➢ Feature Engineering ➢ Rare Location Handling ➢ One-Hot Encoding ➢Train-Test Split	
IU2341050105	Kenil Patel	➢ Feature Scaling ➢ Linear Regression Model ➢ Decision Tree Model ➢ Model Evaluation	
IU2341050114	Vishnu Jikadara	➢ Random Forest Model ➢ Model Comparison ➢ Best Model Selection ➢ Model Saving (.pkl)	

Project Objective:

To develop a House Price Prediction System using Machine Learning techniques and identify the best-performing model for accurate house price prediction using engineered features.

Monthly Work Completed

- Feature engineering performed.
- Rare locations grouped into other.
- One-Hot Encoding applied to categorical features.
- Feature Scaling performed for Linear Regression.
- Dataset split into training and testing sets.
- Trained Linear Regression model.
- Trained Decision Tree Regression model.
- Trained Random Forest Regression model.
- Evaluated models using MAE, RMSE, and R^2 Score.
- Compared model performance.
- Selected the best-performing model.
- Saved trained model (best_ml_model.pkl).
- Saved preprocessing scaler (scaler.pkl).

Challenges Faced

- Handling high-cardinality location features.
- Selecting the most suitable evaluation metrics for regression.
- Preventing overfitting in tree-based models.
- Comparing multiple machine learning models fairly.

Next Month Plan

- Prepare dataset for Deep Learning.
- Build Artificial Neural Network (ANN).
- Train and tune the ANN model.
- Compare ANN performance with ML models.
- Select the best overall model for deployment.

Guide Remarks

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Overall Progress : 40%

Student Signatures:

- Ravi Patel : _____
- Kenil Patel : _____
- Vishnu Jikadara: _____

Project Guide: _____